

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Graphical Transformations

Example (Describe the transformation):

$y = -(x - 2)^2 + 5$ starts from $y = x^2$, moves **right 2**, **up 5**, and **reflects over the x-axis** (because of the $-$).

Directions: Name the parent and describe every shift, stretch, or reflection.

1. $y = (x - 4)^2$

4. $y = |x + 2| - 1$

2. $y = x^2 + 3$

5. $y = \sqrt{x - 5}$

3. $y = -x^2$

Challenge Problems

6. $y = -2|x - 1| + 4$ — describe all transformations.

Answer Key and Procedure

Procedure

1. Identify the parent function.
2. Rewrite as $a f(x - h) + k$.
3. Read h (opposite sign), k , and the sign/size of a .

Answers

1. Quadratic; right 4
2. Quadratic; up 3
3. Quadratic; reflect over the x-axis
4. Absolute value; left 2, down 1
5. Square root; right 5
6. Absolute value; left 1, reflect over x-axis, vertical stretch by 2, up 4